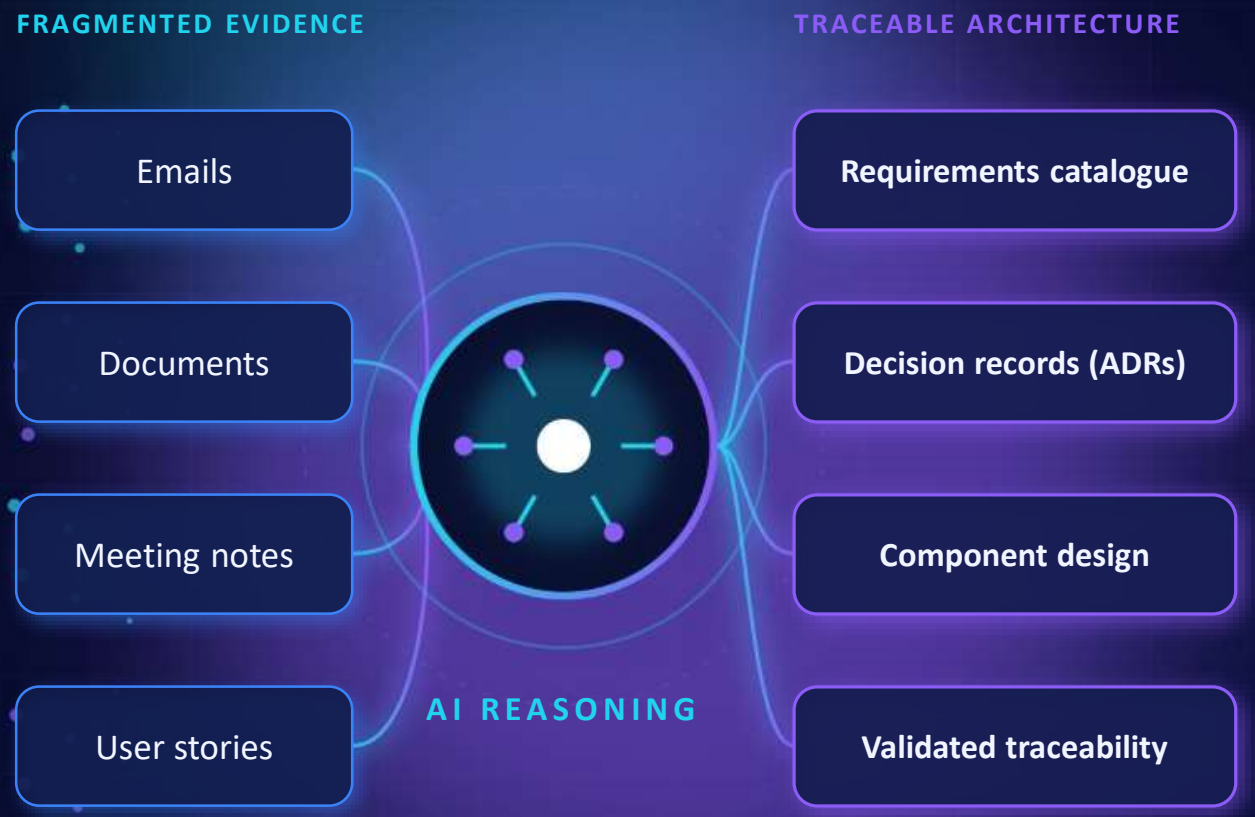
CHALLENGE 02 · THEME 1: AI FOR TECH · DOMAIN: SOFTWARE ENGINEERING

ARCHTRACE

AI Software Architect

Requirements-to-Architecture Agent

Extract · Challenge · Clarify · Decide · Validate



01 Fragmented requirements
Scattered across many sources

02 Conflicting stakeholders
Teams expect different things

03 Ambiguous requirements
Open to multiple interpretations

04 No decision traceability
Why was this design chosen?

How ArchTrace works

Eight stages from raw evidence to validated architecture

KEY INNOVATION · STAGES 4–5



Evidence Ingestion
Normalised evidence items

Requirement Extraction
Candidate requirements with source links

Requirement Classification
FR / NFR / constraint tags

Ambiguity & Conflict Detection
Flagged gaps and contradictions

Clarification Questions
Targeted questions, not guesses

Architecture Decision Records
Drivers, options, trade-offs

Architecture Generation
Components, services, data flow

Validation & Traceability
Coverage and consistency checks

AI that understands before it designs

ILLUSTRATIVE EXAMPLE · NOT A TEST RESULT

BEFORE · STAKEHOLDER EVIDENCE

PRODUCT OWNER

2 s

“The dashboard must refresh every 2 seconds.”

Evidence: stakeholder email

PLATFORM TEAM

30 s

“The backend supports refresh every 30 seconds.”

Evidence: platform constraints document



AFTER · ARCHTRACE ANALYSIS

ARCHTRACE · REQUIREMENT ANALYSIS

ILLUSTRATIVE

CONTRADICTION DETECTED

Refresh interval: 2 s required vs 30 s supported (NFR vs technical constraint)

WHY IT CONFLICTS

The target is 15× faster than the backend currently supports

CLARIFICATION QUESTION

Is 2-second refresh essential for all data, or only critical views? Can the backend limit change?

FLAGGED ASSUMPTION · STATUS

Assume 30 s until confirmed

OPEN

Inside ArchTrace

Technical architecture · stack from the project, arrows show intended data flow



Every decision has a trail

ILLUSTRATIVE IDs · PROPOSED OUTPUTS · MOCK ADR



EVIDENCE SOURCE	REQUIREMENT ID	ADR	COMPONENT	VALIDATION
Stakeholder email	REQ-012	ADR-003	Dashboard service	Coverage check

ADR-003

MOCK

Dashboard refresh strategy

STATUS	Proposed, pending clarification
DRIVERS	Freshness (NFR) vs 30 s backend limit
OPTIONS	Polling · SSE · WebSockets
TRADE-OFFS	Freshness vs complexity and load
OPEN QUESTION	**Is 2 s needed for all views?**

ARCHITECTURE OUTPUTS

PROPOSED

01 Consolidated requirements catalogue

02 Unresolved clarification questions

03 Architecture diagram

04 Component and service decomposition

05 API and data-flow recommendations

06 Architecture Decision Records

07 Architecture risk assessment

Why ArchTrace matters

01

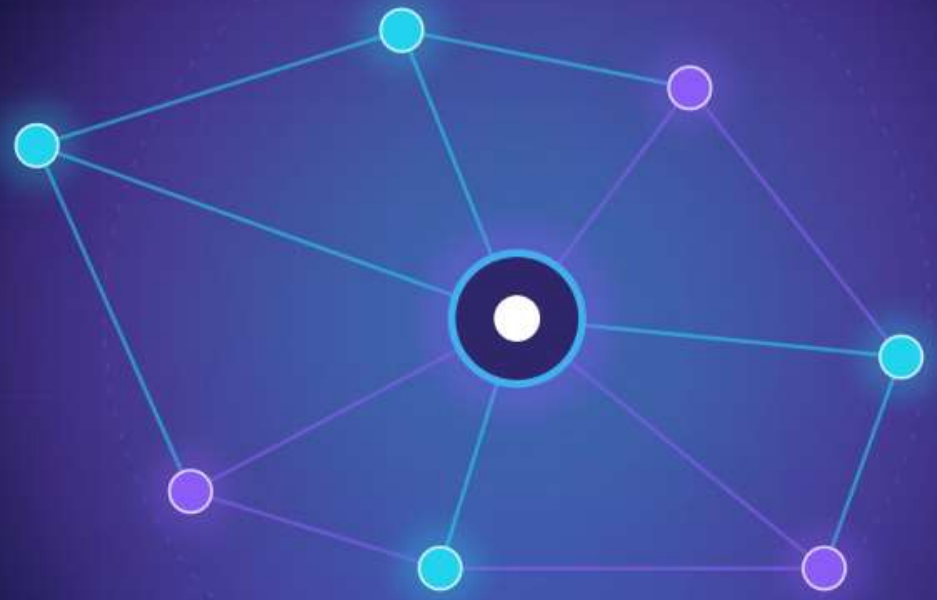
Evidence-based architecture

Every requirement links to its source

02

Intelligent conflict detection

Contradictions surfaced before design



03

Explainable architecture decisions

ADRs record drivers, options, trade-offs

04

End-to-end traceability

Evidence to component to validation

ArchTrace doesn’t just generate architecture.

It explains why that architecture should exist.

VALIDATION DIMENSIONS · DESIGN TARGETS, NO MEASURED RESULTS: Requirement coverage · Contradiction detection · NFR coverage · Architecture consistency · Security · Change adaptability